

Segment Name	Format	Calculation Formula or Explanation
Synchronization Header	Uint16*4	0x0102,0x0304,0x0506,0x0708
Software Version	Uint32	
Total Packet Length	Uint32	Total number of bytes, including the frame header
Hardware Version	Uint32	
Frame Number	Uint32	Current frame number
CPU Time	Uint32	CPU Time
Number of Point Cloud	Uint32	Point cloud number
Output Structure Count	Uint32	Number of output structures
Subframe Number	Uint32	0
If the Output Structure Count > 0, then the following structure exists		
Struct Type	Uint32	1
Struct Length	Uint32	16 * Number of Targets
The following four lines are repeated for each target		
X	float	X (Unit: m)
Y	float	Y (Unit: m)
Z	float	Z (Unit: m)
V	float	Velocity (unit: m/s)
If the total length is not a multiple of 32, add 0x0f at the end for padding		

Analysis Example:

```
02 01 04 03 06 05 08 07 12 08 40 61 60 00 00 00 43 68 0A 00 A7 00 00 00 60 33 A6 C7 03 00 00
00 01 00 00 00 00 00 00 01 00 00 00 30 00 00 00 AD 2C A4 3D 83 51 B6 3D A1 D5 DF BF 00 00
00 00 62 23 58 3E 0C 51 3E 3F 0F CF 0A 40 00 00 00 00 17 7C 87 3F A5 BB 0C 40 8F EF BC 3F 00
00 00 00
```

Synchronization Header:

```
02 01 04 03 06 05 08 07 : 0x0102, 0x0304, 0x0506, 0x0708 (Protocol Compliant)
```

Software Version:

```
12 08 40 61 : 0x61400812
```

Total Packet Length:

```
60 00 00 00 : 0x00000060 = 96 bytes
```

Hardware Version:

```
43 68 0A 00 : 0x000A6843
```

Frame Number:

```
A7 00 00 00 : 0x000000A7 = 167
```

CPU Time (Frequency 600MHz):

```
60 33 A6 C7 : 0xC7A63360 = 3,347,160,416
```

Detected Point Clouds:

03 00 00 00 : 0x00000003 = 3 (Number of Point Clouds)

Output Structure Count:

01 00 00 00 : 0x00000001 = 1

Subframe Number:

00 00 00 00 : 0x00000000 = 0

Structure Type:

01 00 00 00 : 0x00000001 = 1 (Point Cloud Data)

Structure Length:

30 00 00 00 : 0x00000030 = 48 bytes (16 * 3 = 48)

Point Cloud Data (3 point clouds, each 16 bytes, including X, Y, Z, V):

Point Cloud 1:

X: AD 2C A4 3D → 0x3DA42CAD → 0.08016333729028702 m

Y: 83 51 B6 3D → 0x3DB65183 → 0.08902265876531601 m

Z: A1 D5 DF BF → 0xBFDFD5A1 → -1.7487069368362427 m

V: 00 00 00 00 → 0x00000000 → 0 m/s

Point Cloud 2:

X: 62 23 58 3E → 0x3E582362 → 0.21107247471809387 m

Y: 0C 51 3E 3F → 0x3F3E510C → 0.7434241771697998 m

Z: 0F CF 0A 40 → 0x400ACF0F → 2.1688878536224365 m

V: 00 00 00 00 → 0x00000000 → 0 m/s

Point Cloud 3:

X: 17 7C 87 3F → 0x3F877C17 → 1.0584744215011597 m

Y: A5 BB 0C 40 → 0x400CBBA5 → 2.1989529132843018 m

Z: 8F EF BC 3F → 0x3FBCFE8F → 1.4760607481002808 m

V: 00 00 00 00 → 0x00000000 → 0 m/s

Padding: None:

The total length is 96 bytes, which is a multiple of 32, so no padding is required.